

Experiment 6: Observation of electrical resonance in RLC circuit

1. Objectives:

- 1) To understand the concept of resonance in simple RLC circuit.
- 2) To determine the resonance frequency in a parallel RLC circuit and compare this to the expected resonance value.
- 3) To understand concepts related to resonance such as bandwidth and Q- factor.

2. Background:

Resonance is the tendency of a system to oscillate at maximum amplitude when excited at its natural frequency. Electrical resonance occurs when the impedance of part of the circuit reaches a maximum or a minimum at a particular frequency. This is called resonant frequency and its value depends on the circuit elements involved.

The reactance of a capacitor is measured in ohms, is given by $X_C = \frac{1}{2\pi fC}$ and the reactance of an inductor is measured in ohms, given by $X_L = 2\pi fL$.

At low frequencies, the inductor carries most of the current because its impedance $Z_L = j\omega L$ is small. At high frequencies, the capacitor carries most of the current because its impedance $Z_C = \frac{1}{j\omega C}$, is small.

At some point between, when $|Z_L| = |Z_C|$, you might expect that the inductor and the capacitor would each carry half the current. However, this is not the case because we are now dealing with vector quantities. For this condition there is in fact a resonance.

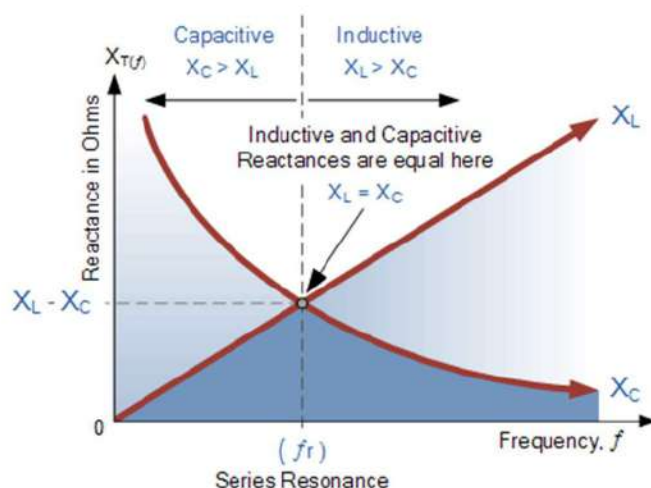


Figure 1. Graphical representation of Impedance and Resonance condition

The condition $|Z_L| = |Z_C|$ can be written as

$$\omega L = \frac{1}{\omega C} \text{ or } \omega = \frac{1}{\sqrt{LC}} \text{ and } f = \frac{1}{2\pi\sqrt{LC}} \quad (1)$$

So the expected resonance frequency is given by equation 1.

$$f_r = \frac{1}{2\pi\sqrt{LC}}$$

Quality Factor

In RLC Circuit, the ratio of resonance frequency to the difference of its neighboring frequencies so that their corresponding signal is $1/\sqrt{2}$ times of the peak value, is called Q-factor of the circuit.

$$\Delta f = f_2 - f_1$$

$$Q = \frac{f_r}{\Delta f}$$

Where f_r is the resonant frequency, and Δf the bandwidth, is the width of the range of frequencies for which the energy is at least $1/\sqrt{2}$ its peak value and Q is known as Quality Factor.

Q is measured as the “sharpness” of the resonance. For instance, when Q is large, the peak of the graph is sharp and the bandwidth is small.

For a parallel RLC circuit, assuming L and C are ideal, the Q factor is given theoretically by the equation.

$$Q = R \sqrt{\frac{C}{L}}$$

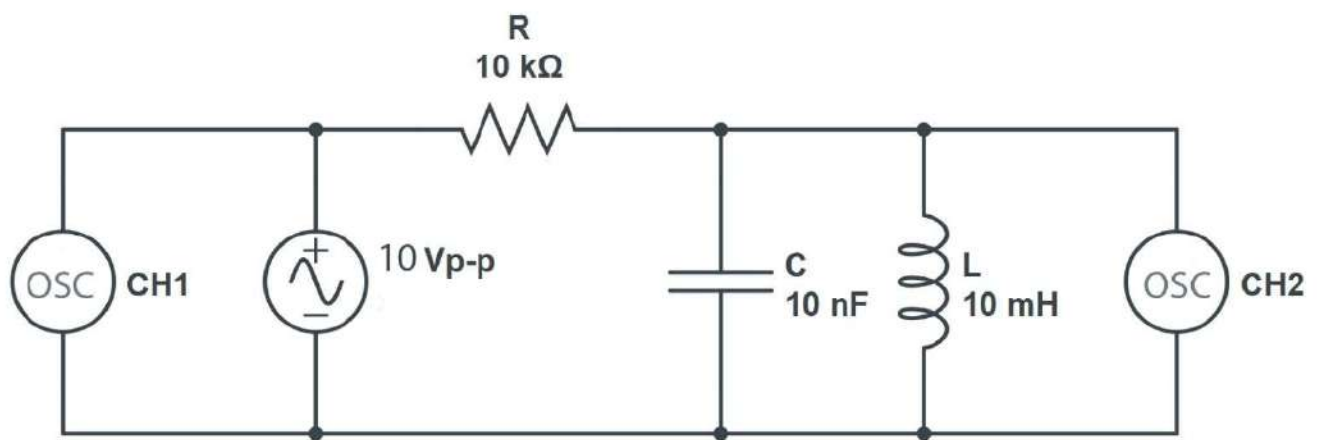


Figure 2: RLC circuit

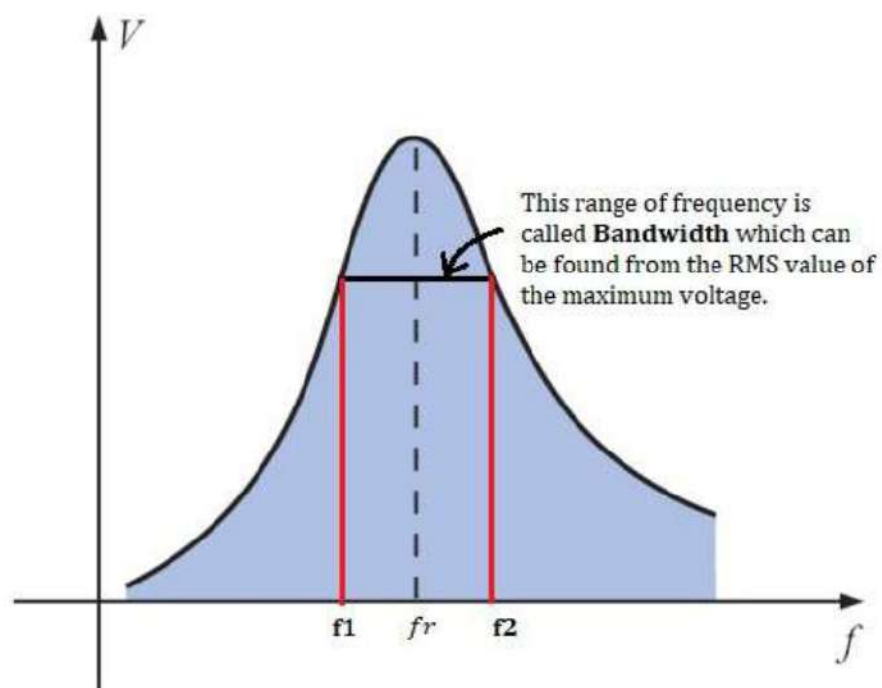


Figure 3: A Graphical representation of resonance in RLC circuit

3. Procedures and Observations:

- a. Before you connect, the circuit to the function generator set the frequency to 1 kHz. Then, using the voltmeter set the function generator's output to 10 volts.
- b. Connect L and C in parallel with the function generator.
- c. Connect the oscilloscope's channel 1 to the signal generator, choose sine wave as an input signal, connect the oscilloscope's channel 2 to the capacitor and inductor, and observe the output signal on the oscilloscope.
- d. Record the values of R, L, and C for this circuit in the space provided in the data section. Use equation 1 to compute the expected resonance frequency and record your result in data table 2.
- e. Record the peak-to-peak voltage from the oscilloscope in data table 1. Now change the function generator frequency to 2 kHz and record the voltage. Then again adjust the frequency to 3 kHz and record the voltage. Continue adjusting the input frequency to each value below the expected resonance frequency computed in step 'd'. Record the voltage for each of these values.
- f. Determine an experimental value for resonance frequency by finding the frequency that produces the largest voltage on the oscilloscope. Record this frequency and voltage.

Lab Report:

Date:	
Name of the Students and IDs	(1)
	(2)
	(3)

Data Tables:

Circuit Parameters:

R= _____

C= _____

L= _____

Table 1

Frequency (KHz)	Output Voltage (V) (pk-pk)	Frequency (KHz)	Output Voltage (V) (pk-pk)
1		16	
2		17	
3		18	
4		19	
5		20	
6		21	
7		22	
8		23	
9		24	
10		25	
11		26	
12		27	
13		28	
14		29	
15		30	

Maximum voltage=

Resonant Frequency, f_r =

Table 2

Calculated Resonant Frequency	
Experimental Resonant Frequency	
% of error	

Tasks and Questions:

#1: Use data obtained in Table 1 to plot output voltage vs frequency graph. Also, label the axes properly.

#2: Find the frequencies f_1 and f_2 and calculate the bandwidth.

#3: Quality Factor (measured from graph) =

#4: Quality Factor (theoretical) =

#5: Compare the theoretical value of Q factor with the experimental value. Find the % error.

Results:

Discussion: